ASMI BASNET

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PROFESSIONAL SUMMARY

AI Engineer and Applied Systems Specialist (M.S. in Analytics candidate, Saint Louis University; B.E. in Electronics & Communication) with 3+ years of experience engineering production LLM applications, autonomous agent workflows, and enterprise automation systems. Proficient in Python, C/C++, TypeScript, and SQL, with proven experience building retrieval-augmented generation (RAG) suites, tool-calling agents (Agno, WebMCP), and predictive forecasting engines. Background architecting air-gapped document intelligence with automated citation verification (PolicyGuard), time-series budget and demand forecasting (Prophet + LSTM), and enterprise data pipelines across Microsoft 365, SharePoint, and REST APIs. Adept at collaborating directly with cross-functional business leaders to translate operational workflows into secure, high-impact software solutions with strict data privacy.

TECHNICAL SKILLS & CORE COMPETENCIES

AI Engineering & LLMs: Retrieval-Augmented Generation (RAG), OpenAI GPT-4o API, Anthropic Claude API (Prompt Caching & Tools), Local LLM Inference (Ollama: Qwen 2.5, Llama 3, Phi-3), Agno Multi-Agent Framework, Model Context Protocol (WebMCP), Vector Databases (ChromaDB, FAISS), Citation Verification, Hallucination Guardrails, Context Window Optimization.

Programming & Custom Software: Python (FastAPI, Flask, Requests, Pandas, NumPy, Scikit-Learn, PyTorch, SQLAlchemy), JavaScript / TypeScript, C/C++, Go (Golang), SQL (PostgreSQL, DuckDB, Snowflake, SQL Server: CTEs, Window Functions), RESTful APIs, Webhooks, Linux CLI, Git / GitHub, Docker, Docker Compose, CI/CD Actions.

Operations & Systems: Workflow Automation, Enterprise Systems Integration (REST APIs, Webhooks, SharePoint / Microsoft 365 Graph), Automated Multi-Tab Excel Reporting (OpenPyXL, Power Query), Time-Series Forecasting (Prophet, LSTM), Anomaly Detection, Operational Dashboarding.

Data Systems, QA & Governance: Automated Data Quality Assurance (IV&V), Schema Drift Monitoring, Audit Logging, Responsible AI & Data Privacy (Air-Gapped / Zero-Leakage Architectures), Star-Schema Dimensional Modeling, Business Intelligence (Microsoft Power BI, DAX, Tableau).

Project Delivery & Collaboration: Agile / Scrum Sprint Planning, JIRA / Project Tracking, Technical Runbooks & API Documentation, Standard Operating Procedures (SOPs), End-User Training & Knowledge-Sharing Sessions, Tool Adoption Telemetry & Stakeholder Engagement.

PROFESSIONAL EXPERIENCE

Saint Louis University — **St. Louis, MO**

Graduate AI Engineer & Research Assistant (SPS-Corps) — *August 2025 – Present*

- Architected and deployed an air-gapped, offline RAG policy compliance suite in Python (FastAPI, ChromaDB, Ollama) that runs local quantized LLMs to analyze confidential enterprise documentation with zero third-party API exposure and zero data leakage.
- Engineered a chain-of-thought verification engine that evaluates citations sentence-by-sentence, flags contradictory requirements (FLAG_CONFLICT), and requests missing context (CLARIFY), aggregating findings into a governance health score to automate contract and policy audits.
- Designed cooperating multi-agent automation workflows leveraging the Agno framework and Anthropic Claude / OpenAI APIs, integrating custom tool-calling routines and REST API connectors that automated cross-departmental research and data synthesis, cutting manual compilation cycles by 90%.
- Engineered automated data ingestion, validation, and ETL pipelines across 8+ operational databases using Python and modular SQL CTEs with window functions, standardizing data dictionaries and cutting query execution times by 40%.
- Built a hybrid forecasting and anomaly detection engine pairing Facebook Prophet with PyTorch LSTM neural networks, automatically flagging demand surges and budget variances beyond $\pm 2.5\sigma$ to deliver predictive resource allocation insights to leadership.
- Authored comprehensive technical documentation, architecture blueprints, and API specifications for deployed AI tools, while leading interactive user training workshops and knowledge-sharing sessions for 25+ administrative staff to facilitate frictionless tool adoption.
- Established a structured enhancement request tracking system to collect end-user feedback, prioritize technical debt and feature requests, and iteratively push containerized updates via Docker Compose and GitHub Actions.

Saint Louis University — **St. Louis, MO**

Data & AI Solutions Analyst Intern — *June 2024 – August 2024*

- Built and deployed 6 interactive executive dashboards across Power BI and Tableau backed by star-schema data models, consolidating 15+ operational KPIs to enable leadership to track resource utilization, productivity, and project milestones.
- Engineered automated data profiling, cleaning, and validation scripts in Python (Pandas) and SQL across 10,000+ institutional records, establishing automated webhook alerts that instantly flag schema drift and transactional anomalies.
- Partnered closely with cross-functional administrative directors and technical teams to map business requirements into technical deliverables, managing sprint milestones and delivering weekly progress presentations to executive stakeholders.
- Implemented automated validation checks and audit routines on sensitive records, enforcing strict governance and institutional privacy compliance.

Entegra Sources Pvt. Ltd. (Mayahold Group) — **Remote / Kathmandu, Nepal**

Systems & Data Analyst — *December 2022 – July 2023*

- Extracted, cleaned, and normalized multi-source operational and telemetry datasets (5,000+ records) into standardized 3NF relational schemas using Python (Pandas, SQLAlchemy) and SQL, ensuring complete data consistency across enterprise systems.
- Validated algorithmic logic and automated backend processing scripts in Python and C/C++, implementing robust exception handling, boundary checks, and audit logging to maintain 99.8% pipeline reliability.
- Authored formal technical documentation detailing data dictionaries, data validation rules, and sampling methodologies to establish organizational data governance standards.

FEATURED AI ENGINEERING & AUTOMATION PROJECTS

PolicyGuard — Enterprise Offline RAG & Document Intelligence Suite **Python, Ollama, ChromaDB, FastAPI, Docker**

- Engineered an air-gapped, privacy-first RAG application running entirely on local quantized LLMs via Ollama to analyze confidential enterprise policy documents and contracts with zero data leakage.
- Implemented automated chain-of-thought citation verification and dynamic cosine similarity guardrails, evaluating retrieved context sentence-by-sentence to detect contradictory clauses (FLAG_CONFLICT) and eliminate manual review overhead.
- Containerized the multi-service architecture using Docker Compose, creating an intuitive Streamlit interface for business reviewers to inspect sourced conflicts alongside an aggregated governance health score.

Nepal Disaster Watch — WebMCP Real-Time Autonomous Command Center **TypeScript, WebMCP, REST APIs, Leaflet.js, GeoJSON**

- Architected an AI-driven crisis intelligence command center featuring 11 native WebMCP tools (6 read, 5 write) for OpenAI's WebMCP Challenge, synchronizing 6 live institutional feeds (NDRRMA, river sensors, Copernicus GloFAS) across all 77 districts of Nepal.
- Engineered resilient feed status adapters (live, snapshot, unreachable), allowing autonomous AI agents to dynamically query spatial risks, pan map layers, and prevent silent system failures during critical operations.
- Implemented dynamic sensor data parsing, event-driven status alerting, and diagnostic verification logic, establishing standard protocols for telemetry verification and remote operations monitoring.

Autonomous Multi-Agent Workflow & Research System **Python, Agno Framework, OpenAI GPT-4o / Claude API, Zapier, REST APIs**

- Built a cooperating multi-agent workflow pairing a Market Scout Agent with an Analytical Synthesis Agent to autonomously monitor data feeds, call external APIs, evaluate valuation anomalies, and synthesize structured executive briefs.
- Integrated event-driven Zapier webhooks and automated messaging endpoints to ingest real-time operational triggers and distribute structured summaries, reducing manual information gathering and reporting cycles by 90%.

CareGap — Clinical Decision Support & Explainable AI Risk Platform **Python, DuckDB, Scikit-Learn, XGBoost, SHAP, FAISS, Streamlit**

- Developed a full-stack decision engine querying 34,000+ synthetic EHR records with sub-second latency using DuckDB columnar Parquet storage, training an ensemble classifier (XGBoost, Random Forest, Lasso) for risk stratification.
- Integrated SHAP TreeExplainer for transparent per-record feature attribution alongside a dual-tier RAG assistant featuring strict prompt guardrails to prevent unsafe recommendations.

EDUCATION & ACADEMIC BACKGROUND

Saint Louis University | Master of Science in Analytics (STEM) **St. Louis, MO | Expected May 2026**

Relevant Coursework: Machine Learning, Artificial Intelligence, LLM Prompt Engineering & Agent Architectures, Cloud Data Architecture, Predictive Big Data Analytics, Database Management & Warehousing.

Tribhuvan University | Bachelor of Engineering in Electronics & Communication (STEM) **Kathmandu, Nepal | Conferred 2023**

Relevant Coursework: Microprocessors & Microcontrollers (C/C++), Control Systems & PID Theory, Computer Networks, Digital Signal Processing, Probability & Statistics.

CERTIFICATIONS & ACCREDITATIONS

- Building with Claude & Anthropic API — Anthropic: Prompt Caching, Model Context Protocol (MCP) Tools, Production Evaluation Loops.
- AI Engineer for Data Scientists Associate — DataCamp: PyTorch Deep Learning Architectures, Scikit-Learn ML Pipelines, Model Validation.
- Microsoft Certified: Power BI Data Analyst Associate (PL-300) — Enterprise Star-Schema Data Modeling, Advanced DAX, KPI Dashboards.
- Project Management & Agile Certification — GlobalShala & Saint Louis University: Agile/Scrum Sprint Planning, Milestone Tracking, Stakeholder Communications.